



Definition

Geometry is the branch of mathematics that deals with the properties of shapes and spaces. The term "GEOMETRY" was derived from the Greek words "ge" meaning earth and "metria" meaning measurements.

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Euclid (c. 330 – c. 275 BC) in his best known book in geometry "ELEMENTS" gives more emphasis on Plane Geometry while Archimedes (c. 287 – c/212 BC) contributed so much to Solid Geometry

Basic Postulates of Euclid

Euclid's Elements is based upon five basic postulates or assumptions which are as follows:

1. A unique straight line can be drawn between any two points.
2. Such a line can be extended indefinitely in either direction.
3. A circle can be drawn in a plane using a given point (a center) and a given distance (a radius).
4. All right angles are equal.
5. Given a line and a point not on the line, there exists exactly one line parallel to the original line passing through the given point.

Postulate number 5 is commonly known as the "parallel postulate"

Basic Geometric Elements

1. Point – a dimensionless geometric figure having no properties other than location or place.
2. Line – the shortest distance between any two points. A line is implicitly a straight line. A line segment is a piece of a line with definite endpoints.
3. Angle – the opening between two lines or two planes that meet. The point where the two lines meet is known as the vertex. Angles comes from Latin word "angulus" which means sharp bend.

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Branches of Geometry

1. **Plane Geometry** – deals with the properties of plane figures or geometrical shapes of two dimensions, such as angles, triangles, square, polygons, conic sections, etc.
2. **Solid Geometry** – deals with the properties of geometric shapes of three dimensions, such as prisms, cones, spheres, pyramids, cylinders, etc.
3. **Euclidean Geometry** – a geometry that is based on the assumptions of Euclid, who about 300 B.C. collected the mathematical knowledge of that time in the 13-volume "Elements". The problems of this type of geometry can be solved by logical reasoning from an initial core of postulates (axioms), a method commonly referred to as the classical axiomatic method of Euclid.
4. **Non-Euclidean Geometry** – a geometry that is not based on the assumption of Euclid.
5. **Projective Geometry** – deals with the study of those properties of plane figures that are unchanged when a given set of points is projected into a second plane.
6. **Trigonometry** – a geometry which specializes on the study of triangles. This subject is divided into two branches namely, Plane and Spherical Trigonometry.
7. **Analytic Geometry** – a geometry which deals with geometric problems by using the coordinates systems and transforming them into algebraic problems. This type of geometry was invented by Rene Descartes. Plane analytic geometry deals primarily with the analysis of equations in two variables while Solid analytic geometry deals with equations in three variables.
8. **Differential Geometry** – a geometry that applies differential and integral calculus to curves, surfaces and other geometric entities.

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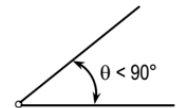
Types of Angles

(According to their measure)

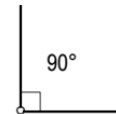
1. **Null or Zero angle** – equal to 0



2. **Acute angle** – greater than 0 but less than 90° or $\pi/2$ radians

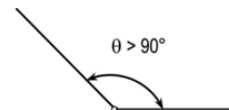


3. **Right angle** – equal to 90° or $\pi/2$ radians



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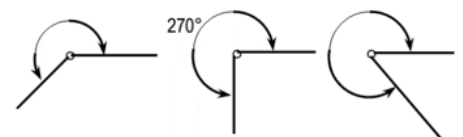
4. **Obtuse angle** – greater than 90° or $\pi/2$ radians but less than 180° or π radians



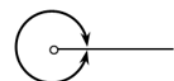
5. **Straight angle** – equal to 180° or π radian



6. **Reflex angle** – greater than 180° or π radians but less than 360° or 2π radians



7. **Full angles or Perigon** – equal to 360° or 2 radians



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